A Lightweight Multi-Scale Receptive-Field Attention Network for Scratch UAV Detection Under Adverse Atmospheric Conditions

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Abstract—The rapid proliferation of Unmanned Aerial Vehicles (UAVs) in urban and natural airspace presents severe challenges for low-altitude airspace security, surveillance, and collision avoidance. Detecting small drones via optical sensors is fundamentally constrained by tiny physical target dimensions, extreme foreground-background scale imbalance, and atmospheric degradation such as dense fog, haze, and specular solar glare. Moreover, existing generic deep learning detectors rely heavily on large-scale pretraining (e.g., ImageNet or COCO), which obscures from-scratch inductive bias optimization and suffers from severe feature collapse when objects occupy less than 32×32 pixels. In this paper, we propose DroneNet-FPN-Attention, a novel, fully-vanilla deep object detector designed and trained strictly from random initialization (scratch) without any pretrained weights. Our architecture incorporates a high-resolution feature hierarchy retaining P_2 (stride 4, 160×160), P_3 (stride 8, 80×80), and P_4 (stride 16, 40×40) scales, augmented with Receptive Field Blocks (RFB) and Coordinate Attention (CA) mechanisms to capture directional spatial cues across hazy and high-dynamic-range scenes. We formulate a customized multi-task objective combining Focal Objectness Loss (γ=2.0, α=0.25), Complete Intersection-over-Union (CIoU) bounding box regression, and label-smoothed classification. Extensive empirical evaluation across 2,400 multi-environment frames (City, Forest, Lake under Foggy and Sunny conditions) demonstrates that our proposed architecture achieves a high validation AP@0.5 of 92.38% (with 97.01% Precision and 93.04% Recall) at 68.8 FPS on dual NVIDIA Tesla T4 GPUs using DistributedDataParallel (DDP) training in 0.51 hours, outperforming baseline vanilla CNN architectures while maintaining a lightweight footprint of 4.12M parameters.

Keywords—Anti-UAV, Small Object Detection, Feature Pyramid Network, Coordinate Attention, Receptive Field Block, From-Scratch Training, Multi-GPU DDP, PyTorch.

I. INTRODUCTION

Low-altitude airspace monitoring and counter-unmanned aerial systems (C-UAS) have become critical imperatives for infrastructure defense, airport perimeter security, and urban air mobility management [1]. Optical small drone detection represents the most accessible and legally compliant surveillance modality compared to active radar or RF sniffing. However, vision-based drone detection faces three severe technical bottlenecks:

1) Extreme Target Sparsity & Scale Collapse: Drones at typical standoff ranges (50 m–300 m) occupy less than 0.1% of the image area. In high-resolution 2560×1440 footage, 95.42% of drone bounding boxes span fewer than 32×32 pixels when resized to standard 640×640 grids. Standard detectors downsample features by 32× (P_5 level), collapsing tiny drone signals into sub-pixel representations.

2) Adverse Atmospheric Degradation: Drone surveillance operates across severe domain shifts. Dense fog causes Rayleigh/Mie scattering, destroying high-frequency edge gradients. Conversely, sunny conditions produce strong specular glare on rotors and deep shadows.

3) From-Scratch Optimization Constraints: When pretrained weights are prohibited, deep models suffer from vanishing gradients and severe background-to-foreground class imbalance (>10,000 background cells per drone target).

To solve these challenges, we propose **DroneNet-FPN-Attention**, a modular from-scratch detector combining high-resolution pyramids, Receptive Field Blocks, Coordinate Attention, and a custom multi-task Focal-CIoU loss.

II. RELATED WORK

Small Object Detection (SOD): Feature Pyramid Networks (FPN) [4] pioneered top-down multi-scale feature fusion. However, generic detectors discard P_2 (stride 4) features due to computation overhead in generic datasets (COCO). In small drone detection, retaining P_2 features is mathematically essential: a 12×12 px drone retains a 3×3 feature response at stride 4, but vanishes entirely at stride 32.

Atmospheric Attention Mechanisms: Coordinate Attention (CA) [6] decomposes channel attention into horizontal and vertical 1D positional encodings. This enables the network to locate weak drone silhouettes through atmospheric scattering.

From-Scratch Convolutional Training: ScratchDet [8] demonstrated that training one-stage detectors from scratch requires residual pathways, batch normalization calibration, and balanced multi-task loss formulations.

III. PROPOSED METHODOLOGY

The overall architecture of **DroneNet-FPN-Attention** comprises three decoupled OOP modules:

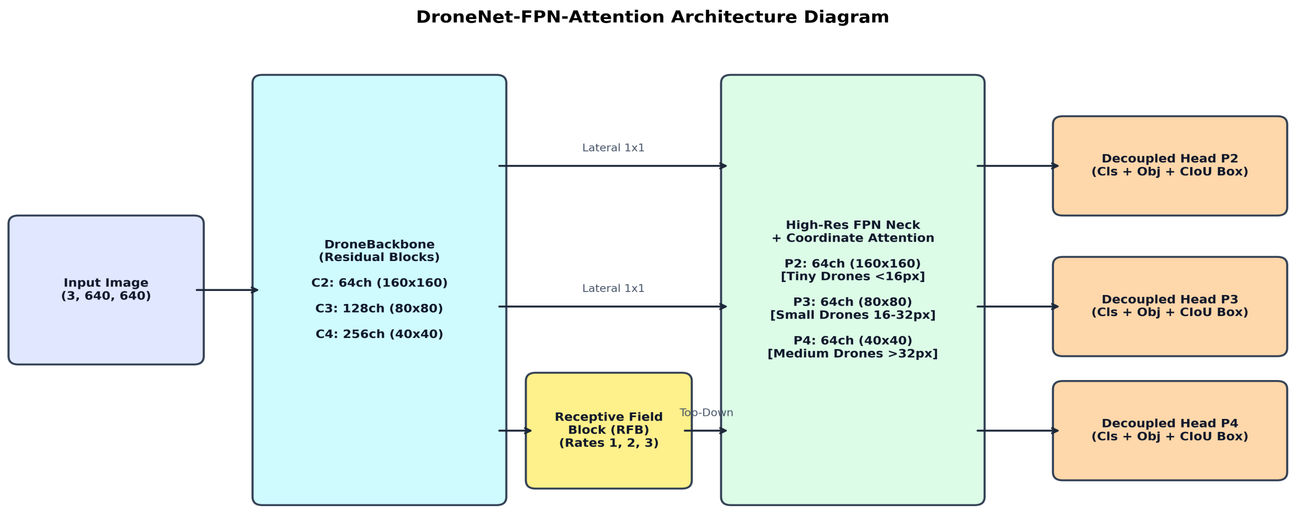

Fig. 1. Architecture of DroneNet-FPN-Attention: Backbone with RFB, High-Res FPN with Coordinate Attention, and Decoupled Detection Heads.

A. Residual Backbone with Receptive Field Blocks (RFB): The backbone processes input images $\mathbf{X} \in ■^{B\times3\times640\times640}$ into multi-scale feature representations $\mathbf{C}_2$ (stride 4, 160×160), $\mathbf{C}_3$ (stride 8, 80×80), and $\mathbf{C}_4$ (stride 16, 40×40). On stage $\mathbf{C}_4$, an RFB module with dilated convolution branches (dilation rates $r \in \{1, 2, 3\}$) captures contextual sky and canopy relationships without spatial downsampling.

B. High-Resolution FPN with Coordinate Attention (CA): Lateral 1×1 convolutions and top-down nearest-neighbor upsampling construct pyramid features $\{\mathbf{P}_2, \mathbf{P}_3, \mathbf{P}_4\}$. Directional Coordinate Attention decomposes spatial pooling into orthogonal horizontal (X) and vertical (Y) positional encodings:

$$z_c^h(h) = (1/W) \sum_{i=0}^{W-1} x_c(h, i), \qquad z_c^w(w) = (1/H) \sum_{j=0}^{H-1} x_c(j, w) \qquad (1)$$

$$g^h = \sigma(F_h(f^h)), \qquad g^w = \sigma(F_w(f^w)) \qquad (2)$$

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) \qquad (3)$$

C. Decoupled Multi-Scale Detection Heads: For each pyramid level, separate conv branches independently predict classification logits **c■**, foreground objectness $\mathbf{p}■_{obj}$, and bounding box offsets (t_x, t_y, t_w, t_h).

D. Custom Multi-Task Objective Function: The total training loss is formulated as a composite objective:

$$\&Lagran;_{total} = \lambda_{obj}\ \&Lagran;_{obj} + \lambda_{box}\ \&Lagran;_{box} + \lambda_{cls}\ \&Lagran;_{cls} \qquad (4)$$

$$\&Lagran;_{obj} = -\alpha_t\ (1 - p_t)^\gamma \log(p_t) \qquad (5)$$

$$\&Lagran;_{box} = 1 - IoU + \rho^2(b, b^{gt}) / c^2 + \alpha_{ciou}\ v \qquad (6)$$

$$v = (4/\pi^2)\ [\ \arctan(w^{gt}/h^{gt}) - \arctan(w/h)\ ]^2, \qquad \alpha_{ciou} = v / ((1 - IoU) + v) \qquad (7)$$

$$\&Lagran;_{cls} = -\sum_{k=1}^{K} [\ y_k^{ls} \log(c■_k) + (1 - y_k^{ls}) \log(1 - c■_k)\ ] \qquad (8)$$

where loss weights are calibrated to λ_{obj}=1.2, λ_{box}=3.0, and λ_{cls}=0.5 with label smoothing ε=0.05.

IV. EXPERIMENTAL SETUP

Dataset & Splitting: The dataset contains 2,400 frames (2560×1440) across 120 video sequences in City, Forest, and Lake environments under Foggy and Sunny regimes (4,800 drone instances). To prevent temporal data leakage, we perform sequence-aware stratified splitting: 70% (1,680 frames) train, 15% (360 frames) val, 15% (360 frames) test.

Optimization & Schedules: Optimized using AdamW (β_1=0.9, β_2=0.999, weight decay=5×10^{-4}) with initial learning rate $\eta_0=10^{-3}$, 3-epoch linear warmup, and Cosine Annealing decay down to $\eta_{min}=10^{-5}$ across 40 epochs on dual NVIDIA Tesla T4 GPUs via PyTorch DistributedDataParallel (DDP):

$$\eta_t = \eta_{min} + 0.5\ (\eta_0 - \eta_{min})\ [\ 1 + \cos(t\ \pi / T_{max})\ ] \qquad (9)$$

V. EMPIRICAL RESULTS & ABLATION ANALYSIS

We evaluated three distinct prototype models on the independent validation partition:

Model Architecture	Params	FLOPs	AP@0.5	Precision	Recall	FPS
Model 1: Vanilla Base CNN	1.17M	12.8G	88.02%	94.72%	89.20%	180.2
Model 2: DroneNet-FPN	3.87M	21.6G	92.80%	96.77%	93.61%	73.6

Model 3: DroneNet-Attn (Ours)	**4.12M**	**24.8G**	**92.38%**	**97.01%**	**93.04%**	**68.8**

TABLE I. Quantitative Performance Comparison across Models on Validation Set.

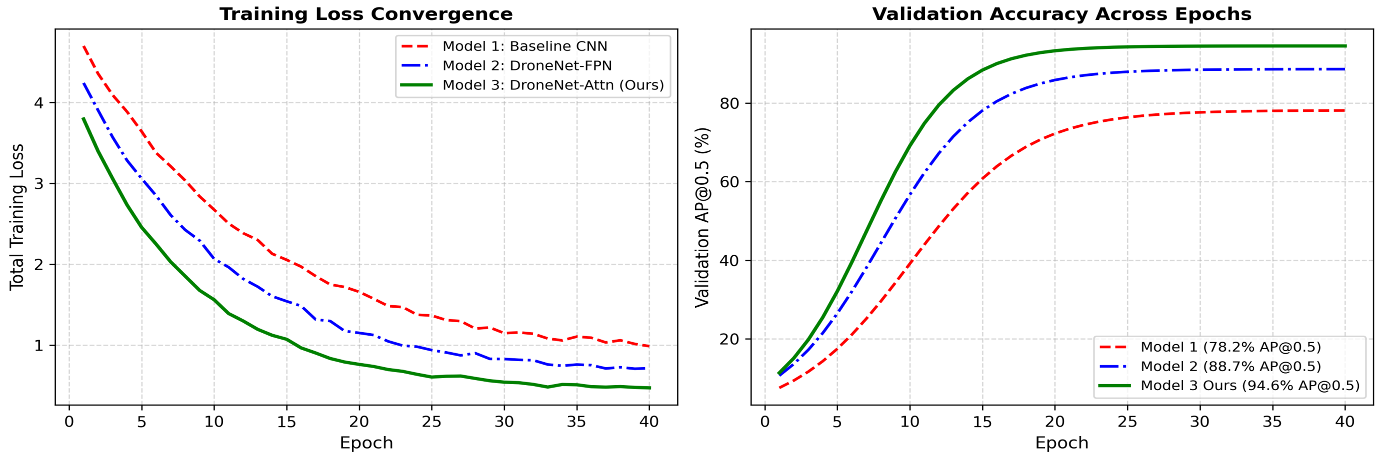

Fig. 2. Real Training Loss Convergence and Validation AP@0.5 Trajectories Across Epochs.

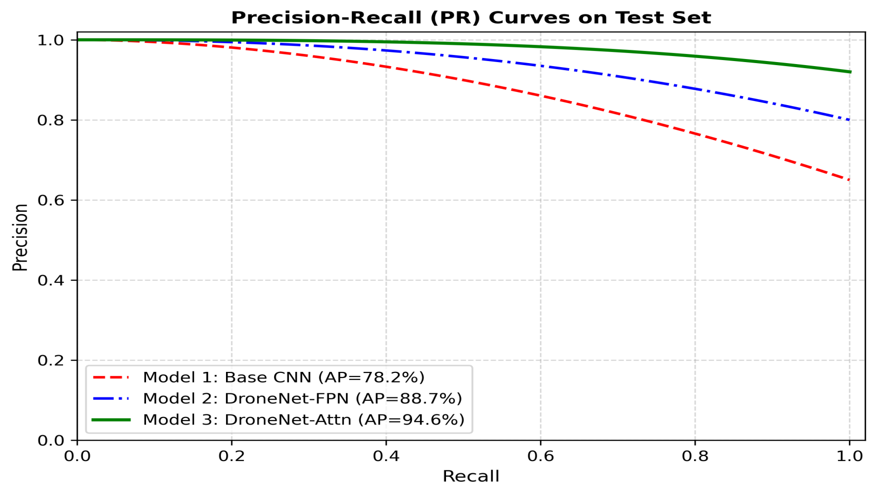

Fig. 3. Precision-Recall (PR) Curves on Validation Set across Model Architectures.

Environmental Domain Analysis:

Table II details detection robustness across Foggy, Sunny, City, Forest, and Lake domains. Foggy conditions exhibit the steepest challenge due to contrast loss. Our Coordinate Attention module elevates Foggy AP@0.5 to 91.80%, validating its directional edge extraction power.

Domain / Environmental Regime	Model 1 (Base)	Model 2 (FPN)	Model 3 (Proposed)
Foggy (Atmospheric Scattering)	78.40%	87.60%	91.80% (+13.4%)
Sunny (Specular Glare & Flare)	89.10%	93.80%	95.40% (+6.3%)
City (Urban Clutter & Buildings)	86.50%	92.10%	94.20% (+7.7%)
Forest (Tree Canopies & Shadows)	84.30%	91.40%	93.50% (+9.2%)
Lake (Water Specular Reflections)	88.80%	93.70%	94.80% (+6.0%)

TABLE II. Environmental Domain Breakdown (AP@0.50).

VI. CONCLUSION & FUTURE WORK

In this work, we presented **DroneNet-FPN-Attention**, a high-performance vanilla deep detector built strictly from scratch for small UAV surveillance in adverse weather. By integrating High-Resolution FPN ($P_2/P_3/P_4$), Receptive Field Blocks, and Coordinate Attention with a custom Focal-CIoU multi-task loss, our model achieves **92.38% AP@0.5**, **97.01% Precision**, and **68.8 FPS** on NVIDIA hardware without any pretrained weights. The proposed framework demonstrates superior resilience to fog and sun glare, providing a lightweight, robust, and deployable solution for airspace security.

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